

IMAP Charge Prediction

IMAP — Integrated Modeling & Analysis Platform

≈9%
of lab test time spent on charge optimization

Product family	Dual Fuel Heat Pump				Standard Efficiency Unit					High Efficiency Unit			
Tonnage	12.5	15	20	25	15	17.5	20	25	27.5	15	17.5	20	23
Charge optimization (days)	17	9	14	17	10	10	4	15	15	8	11	7	10
Test duration (days)	140	138	150	143	150	111	54	141	160	109	133	94	133
Charge optimization % of total	12.1%	6.5%	9.3%	11.9%	6.7%	9.0%	7.4%	10.6%	9.4%	7.3%	8.2%	7.4%	7.5%
Average	10.0%				8.6%					7.6%			

Impact on Dual Fuel Heat Pump test plan

Parameter	Before — accuracy 80-85%				After — accuracy 95-97%			
Tonnage	12.5	15	20	25	12.5	15	20	25
Charge opt. days	17	9	14	17	4	0	0	5
Test duration (days)	140	138	150	143	127	0	0	129
Total days, all units	571				256			

In-between 15 & 20 ton units removed from lab testing — charge predicted by IMAP at 95-97% accuracy.

≈50%

reduction in total test days
571 → 256 days

ROOT CAUSE — IMAP GAP

IMAP charge prediction accuracy today:
~80-85%
Gap to close: 10-13 pts (target 95-97%)

- ✓ More lab slots — higher utilization of test chambers
- ✓ Faster product development cycles
- ✓ Benefit scales to every future unit tested

Avg. % difference vs. lab
Heat Pump unit: 17%
Cooling unit: 20%